DBMS – SQL QUERY IMPLEMENTATION FOR E-COMMERCE DATABASE

Project Title: Big Basket E-commerce Database Management System

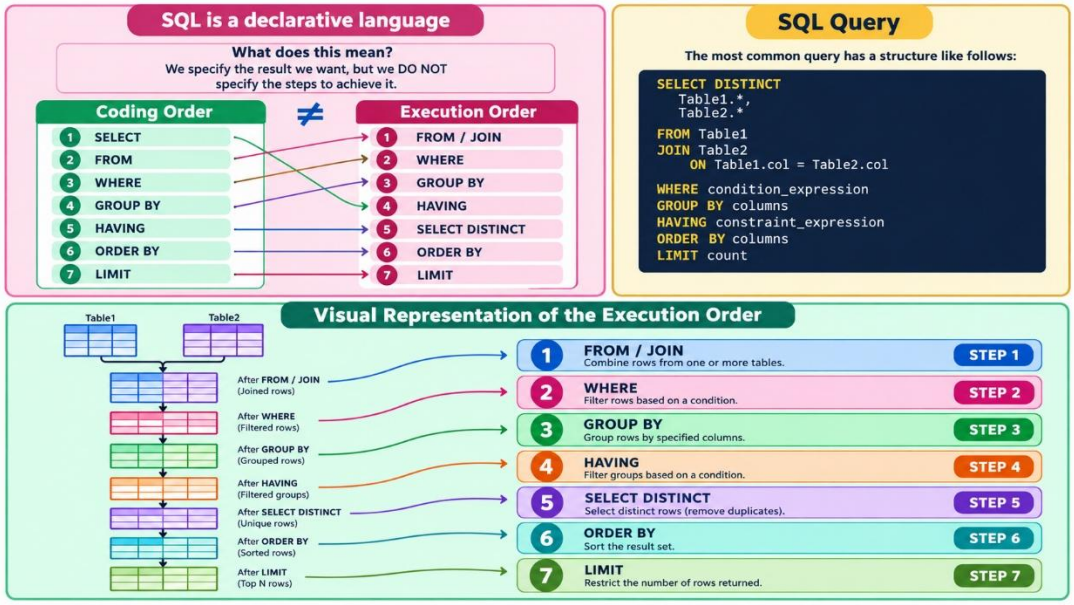
Objectives:

1. Perform **SELECT, WHERE, ORDER BY,** and **DISTINCT** queries.
2. Search products based on price, category and availability.
3. Retrieve customer and product information.
4. Apply filter conditions.
5. Generate a basic business report

1) Basic SQL queries

1.1 SELECT query

The **SELECT** statement is used to retrieve data from a table



Display all products

```
Select * from Product;    Syntax Error at line 1, column 22⏎Select * from Product;⏎ ^^^⏎Expected: sq
PRODUCT_ID PRODUCT_NAME                                        CATEGORY_ID
---------- -------------------------------------------------- -----------
SUPPLIER_ID BRAND                          PRICE      STOCK EXPIRY_DA
----------- ------------------------------ ---------- ---------- ---------
       101 Laptop                                                       1
      1001 Dell                            55000         10

       102 Smartphone                                                   1
      1002 Samsung                         25000         20

       103 Tablet                                                       1
      1003 Lenovo                          18000         15


PRODUCT_ID PRODUCT_NAME                                        CATEGORY_ID
---------- -------------------------------------------------- -----------
SUPPLIER_ID BRAND                          PRICE      STOCK EXPIRY_DA
----------- ------------------------------ ---------- ---------- ---------
       104 Keyboard                                                     1
      1004 HP                               1200         25

       105 Mouse                                                        1
      1005 Logitech                          800         30

       106 Monitor                                                      1
      1006 LG                              15000         12
```

Display selected product details

```
Select product_id,product_name,price
from product;

PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
       101 Laptop                                              55000
       102 Smartphone                                          25000
       103 Tablet                                              18000
       104 Keyboard                                             1200
       105 Mouse                                                 800
       106 Monitor                                             15000
       107 Printer                                             12000
       108 Speaker                                              3500
       109 Webcam                                               2500
       110 Headphones                                           4500
       111 T-Shirt                                              1200

PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
       112 Jeans                                                2500
       113 Formal Shirt                                         1800
       114 Kurti                                                2200
       115 Saree                                                4500
       116 Hoodie                                               2800
       117 Jacket                                               3500
       119 Skirt                                                2000
       120 Sweater                                              2400
       121 DBMS Book                                             700
       122 Python Programming                                    850
```

1.2 WHERE Clause

The WHERE clause is used to filter records based on a condition.

Find products with a price greater than 100

```
Select product_id,product_name,price
from product
where price>100;
PRODUCT_ID PRODUCT_NAME                                             PRICE
---------- -------------------------------------------------- ----------
       101 Laptop                                                  55000
       102 Smartphone                                              25000
       103 Tablet                                                  18000
       104 Keyboard                                                 1200
       105 Mouse                                                     800
       106 Monitor                                                 15000
       107 Printer                                                 12000
       108 Speaker                                                  3500
       109 Webcam                                                   2500
       110 Headphones                                               4500
       111 T-Shirt                                                  1200

PRODUCT_ID PRODUCT_NAME                                             PRICE
---------- -------------------------------------------------- ----------
       112 Jeans                                                    2500
       113 Formal Shirt                                             1800
       114 Kurti                                                    2200
       115 Saree                                                    4500
       116 Hoodie                                                   2800
       117 Jacket                                                   3500
       119 Skirt                                                    2000
       120 Sweater                                                  2400
       121 DBMS Book                                                 700
       122 Python Programming                                        850
       123 Java Programming                                          900
```

Find products with a price less than 100

```
Select product_id,product_name,price
from product
where price<100;
PRODUCT_ID PRODUCT_NAME                                             PRICE
---------- -------------------------------------------------- ----------
       144 Sugar                                                      55
       145 Salt                                                       30
       147 Biscuits                                                   40
       150 Noodles                                                    60
       191 Notebook                                                   60
       194 Eraser                                                     10
       195 Sharpener                                                  15
       199 Sticky Notes                                               80

8 rows selected.
```

1.3 ORDER BY

The ORDER BY clause is used to sort the query results.

Products from Lowest to Highest

```
Select product_id,product_name,price
from product
Order by price asc;
```

PRODUCT_ID	PRODUCT_NAME	PRICE
194	Eraser	10
195	Sharpener	15
145	Salt	30
147	Biscuits	40
144	Sugar	55
191	Notebook	60
150	Noodles	60
199	Sticky Notes	80
198	File Folder	100
193	Pencil Box	150
146	Toor Dal	160

PRODUCT_ID	PRODUCT_NAME	PRICE
143	Cooking Oil	180
196	Marker Set	180
174	Face Wash	250
148	Tea	250
197	Geometry Box	250
177	Kajal	300
149	Coffee	300
168	Skipping Rope	300
179	Body Lotion	320
200	A4 Paper	350
175	Moisturizer	400

Products from Highest to Lowest

```
Select product_id,product_name,price
from product
Order by price desc;
```

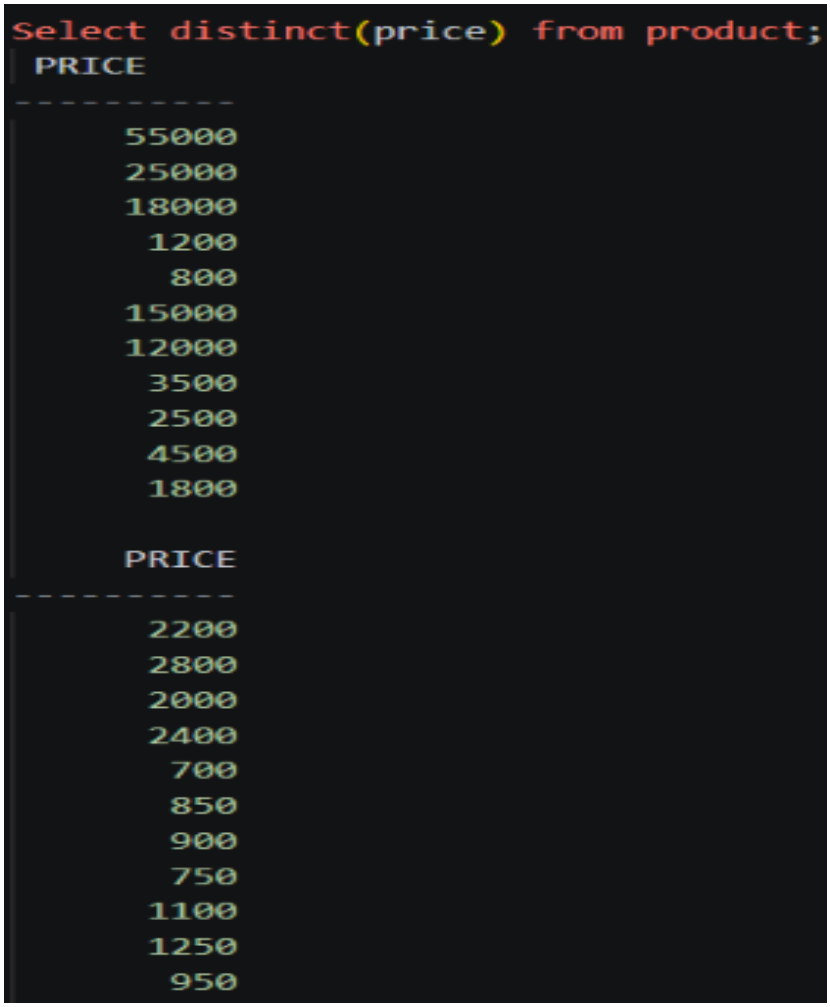
PRODUCT_ID	PRODUCT_NAME	PRICE
101	Laptop	55000
136	Bed	30000
102	Smartphone	25000
134	Sofa	25000
137	Wardrobe	22000
103	Tablet	18000
106	Monitor	15000
133	Dining Table	12000
107	Printer	12000
140	TV Stand	8000
132	Office Chair	7000

PRODUCT_ID	PRODUCT_NAME	PRICE
135	Bookshelf	7000
139	Coffee Table	6500
138	Study Table	5500
110	Headphones	4500
158	Loafers	4500
115	Saree	4500
131	Computer Table	4500
117	Jacket	3500
108	Speaker	3500
164	Tennis Racket	3500
151	Running Shoes	3500

1.4 DISTINCT

The DISTINCT keyword is used to display unique values.

Display the unique categories



2) Search Products

2.1 Search Products Based on Price

This displays products whose price is between ₹100 and ₹400.

```
Select product_id,product_name,price
from product
Where price between 100 and 700;
```

PRODUCT_ID	PRODUCT_NAME	PRICE
121	DBMS Book	700
129	Operating Systems	700
142	Wheat Flour	450
143	Cooking Oil	180
146	Toor Dal	160
148	Tea	250
149	Coffee	300
155	Slippers	600
157	Flip Flops	400
163	Cricket Ball	450
168	Skipping Rope	300

PRODUCT_ID	PRODUCT_NAME	PRICE
171	Face Cream	450
173	Lipstick	599
174	Face Wash	250
175	Moisturizer	400
176	Sunscreen	650
177	Kajal	300
179	Body Lotion	320
180	Compact Powder	550
181	Toy Car	600
186	Puzzle Game	450
189	Rubik Cube	500

2.2 Search Products Based on Category

These displays products belonging to category 10

```
Select Product_id,product_name,Category_id
from product
where category_id =10;
PRODUCT_ID PRODUCT_NAME                                           CATEGORY_ID
---------- -------------------------------------------------- -----------
       191 Notebook                                                    10
       193 Pencil Box                                                  10
       194 Eraser                                                      10
       195 Sharpener                                                   10
       196 Marker Set                                                  10
       197 Geometry Box                                                10
       198 File Folder                                                 10
       199 Sticky Notes                                                10
       200 A4 Paper                                                    10

Select * from Customer;
CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE_NO        DOB
--------------- ---------
ADDRESS
--------------------------------------------------------------------------------
        101 Arun Kumar
arun@gmail.com
9876543210      12-MAY-98
Chennai


CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE_NO        DOB
```



3. Retrieve Customer Information

Display all customers and search for a particular customer

```
Select Customer_ID,Customer_Name,Email,Phone_No
from Customer;
CUSTOMER_ID CUSTOMER_NAME
----------- ----------------------------------------
EMAIL
--------------------------------------------------
PHONE_NO
---------------
         101 Arun Kumar
arun@gmail.com
9876543210

         102 Priya Sharma
priya@gmail.com
9876543211

CUSTOMER_ID CUSTOMER_NAME
----------- ----------------------------------------
EMAIL
--------------------------------------------------
PHONE_NO
---------------

         103 Rahul Singh
rahul@gmail.com
9876543212

         104 Sneha Reddy
sneha@gmail.com
```

Search for a particular customer

```
Select *
from Customer
where customer_name like 'A%';
CUSTOMER_ID CUSTOMER_NAME
----------- ----------------------------------------
EMAIL
--------------------------------------------------
PHONE_NO        DOB
--------------- ---------
ADDRESS
--------------------------------------------------
        | 101 Arun Kumar
arun@gmail.com
9876543210      12-MAY-98
Chennai


CUSTOMER_ID CUSTOMER_NAME
----------- ----------------------------------------
EMAIL
--------------------------------------------------
PHONE_NO        DOB
--------------- ---------
ADDRESS
--------------------------------------------------
        | 106 Anjali Das
anjali@gmail.com
9876543215      25-JAN-05
Kolkata
```

4. Apply Filtering Conditions

Multiple conditions can be combined using AND, OR, and NOT.

4.1 AND Condition

This displays products that satisfy both conditions.

```
Select Product_id,product_name,price
from product
where price<800 and category_id=3;

PRODUCT_ID PRODUCT_NAME                                        PRICE
---------- -------------------------------------------------- ----------
       121 DBMS Book                                             700
       124 Data Structures                                       750
       129 Operating Systems                                     700
```

4.2 OR Condition

This displays products belonging to either category 103 or price greater than 800

```
Select Product_id,product_name,price
from product
where price>800 or category_id=3;
```

PRODUCT_ID	PRODUCT_NAME	PRICE
101	Laptop	55000
102	Smartphone	25000
103	Tablet	18000
104	Keyboard	1200
106	Monitor	15000
107	Printer	12000
108	Speaker	3500
109	Webcam	2500
110	Headphones	4500
111	T-Shirt	1200
112	Jeans	2500

PRODUCT_ID	PRODUCT_NAME	PRICE
113	Formal Shirt	1800
114	Kurti	2200
115	Saree	4500
116	Hoodie	2800
117	Jacket	3500
119	Skirt	2000
120	Sweater	2400
121	DBMS Book	700
122	Python Programming	850
123	Java Programming	900
124	Data Structures	750

4.3 IN Operator

The IN operator is used to check whether a value matches any value in a specified list.

```
Select Product_id,product_name,price
from product
where category_id in (1,2,3);
```

PRODUCT_ID	PRODUCT_NAME	PRICE
101	Laptop	55000
102	Smartphone	25000
103	Tablet	18000
104	Keyboard	1200
105	Mouse	800
106	Monitor	15000
107	Printer	12000
108	Speaker	3500
109	Webcam	2500
110	Headphones	4500
111	T-Shirt	1200

PRODUCT_ID	PRODUCT_NAME	PRICE
112	Jeans	2500
113	Formal Shirt	1800
114	Kurti	2200
115	Saree	4500
116	Hoodie	2800
117	Jacket	3500
119	Skirt	2000
120	Sweater	2400
121	DBMS Book	700
122	Python Programming	850
123	Java Programming	900

4.4 BETWEEN Operator

The BETWEEN operator is used to retrieve records within a specified range.

```
Select Product_id,product_name,price
from product
where price between 1000 and 5000;
```

PRODUCT_ID	PRODUCT_NAME	PRICE
104	Keyboard	1200
108	Speaker	3500
109	Webcam	2500
110	Headphones	4500
111	T-Shirt	1200
112	Jeans	2500
113	Formal Shirt	1800
114	Kurti	2200
115	Saree	4500
116	Hoodie	2800
117	Jacket	3500

PRODUCT_ID	PRODUCT_NAME	PRICE
119	Skirt	2000
120	Sweater	2400
125	Machine Learning	1100
126	Artificial Intelligence	1250
131	Computer Table	4500
151	Running Shoes	3500
152	Sports Sandals	2200
153	Formal Shoes	2800
154	Sneakers	3200
156	Heels	2500
158	Loafers	4500

5. Basic Business Reports

5.1 Product Price Report

Displays product information, including product ID, product name, and price.

```
Select Product_id,product_name,price
from product
Order by price desc;
```

PRODUCT_ID	PRODUCT_NAME	PRICE
104	Keyboard	1200
108	Speaker	3500
109	Webcam	2500
110	Headphones	4500
111	T-Shirt	1200
112	Jeans	2500
113	Formal Shirt	1800
114	Kurti	2200
115	Saree	4500
116	Hoodie	2800
117	Jacket	3500

PRODUCT_ID	PRODUCT_NAME	PRICE
119	Skirt	2000
120	Sweater	2400
125	Machine Learning	1100
126	Artificial Intelligence	1250
131	Computer Table	4500
151	Running Shoes	3500
152	Sports Sandals	2200
153	Formal Shoes	2800
154	Sneakers	3200
156	Heels	2500
158	Loafers	4500

5.2 Customer Order Report

Displays customer details and order information, including customer ID, customer name, order ID, order date, and total amount.

```
Select
c. Customer_ID,
c.Customer_Name,
o.Order_ID,
o.Order_Date,
o.Total_Amount
from Customer c
inner join Orders o
on c.Customer_ID = o.Customer_ID
order by o.Order_Date desc;
CUSTOMER_ID CUSTOMER_NAME                                          ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
        101 Arun Kumar                                               1001
21-SEP-26          900

        102 Priya Sharma                                             1002
20-SEP-26         1250

        109 Rohan Gupta                                              1010
20-SEP-26        28000


CUSTOMER_ID CUSTOMER_NAME                                          ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
        103 Rahul Singh                                              1003
19-SEP-26         1800

        101 Arun Kumar                                               1004
18-SEP-26          750

        104 Sneha Reddy                                              1005
17-SEP-26          500
```

5.3 Product Order Report

Displays product order details, including order ID, product ID, quantity, unit price, and subtotal.

```
Select
od.Order_ID,
od.Product_ID,
od.Quantity,
od.Unit_Price,
od.Subtotal
from Order_Details od
Order by od.Subtotal desc;
```

ORDER_ID	PRODUCT_ID	QUANTITY	PRICE	SUB_TOTAL
1001	101	2	70000	70000
1003	102	1	45100	45000
1006	107	2	16000	32000
1010	110	2	14000	28000
1008	105	2	11000	22000
1004	103	2	9000	18000
1009	109	1	15000	15000
1005	106	1	12500	12500
1001	104	2	5250	10500
1007	108	1	8500	8500
1002	104	1	5000	5000

11 rows selected.

5.4 Customer-wise Total Orders

Displays the total number of orders placed by each customer.

```
SELECT
    c.Customer_ID,
    c.Customer_Name,
    COUNT(o.Order_ID) AS Total_Orders
FROM Customer c
JOIN Orders o
    ON c.Customer_ID = o.Customer_ID
GROUP BY c.Customer_ID, c.Customer_Name
ORDER BY Total_Orders DESC;

CUSTOMER_ID CUSTOMER_NAME                                          TOTAL_ORDERS
----------- -------------------------------------------------- ------------
        101 Arun Kumar                                                    3
        103 Rahul Singh                                                   1
        109 Rohan Gupta                                                   1
        105 Vikram Patel                                                  1
        106 Anjali Das                                                    1
        107 Karan Mehta                                                   1
        102 Priya Sharma                                                  1
        104 Sneha Reddy                                                   1

8 rows selected.
```

5.5 Customer-wise Total Purchase Amount

Displays the total purchase amount for each customer.

```
SELECT
    c.Customer_ID,
    c.Customer_Name,
    SUM(o.Total_Amount) AS Total_Purchase
FROM Customer c
JOIN Orders o
    ON c.Customer_ID = o.Customer_ID
GROUP BY c.Customer_ID, c.Customer_Name
ORDER BY Total_Purchase DESC;

CUSTOMER_ID CUSTOMER_NAME                                        TOTAL_PURCHASE
----------- -------------------------------------------------- --------------
        109 Rohan Gupta                                               28000
        101 Arun Kumar                                                 3150
        107 Karan Mehta                                                2500
        103 Rahul Singh                                                1800
        102 Priya Sharma                                               1250
        106 Anjali Das                                                 1200
        105 Vikram Patel                                                850
        104 Sneha Reddy                                                 500

8 rows selected.
commit;
Commit complete.
```